Zhiying Cui

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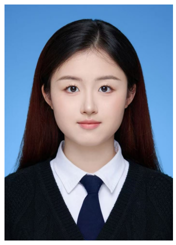

Education

Ningbo University, Computer Science and Technology — Sep. 2022 – Jun. 2026

- **GPA: 3.52/4.00 Average Score: 86.41/100**
- **Major Courses (/100):**
  - Assembly Language and Interface Technique (95)
  - Big Data Processing Techniques (92)
  - Computational Thinking and C Programming (93)
  - Principles and Application of Database Systems (95)
  - Digital Image Processing (91)
  - Probability and Statistics (90)
- **Scholarships:** Provincial Government Scholarship (2024, 2025); Jiang Xinghao First Prize for Student Research Achievements (2025); Loctek Innovation and Entrepreneurship Scholarship, Third Class (2024); University Grand Prize Scholarship (2023)

Research Projects

A Multi-Scale Fusion Approach with Attention Mechanisms for Glass Curtain Wall Damage Detection — Dec. 2024 – Present

- Traditional manual methods for detecting damaged glass curtain walls are inefficient and pose safety risks. To address this, we proposed a method combining YOLOv8's head with Grad-CAM to generate heatmaps for key regions, which were then cropped and processed by a detail detection network. This network uses multi-scale fusion and attention mechanisms to enhance both local and global feature extraction, particularly for damaged glass textures.
- Responsible for designing and implementing the method, achieving 95.4% accuracy, and obtaining one **National Invention Patent** as the **first inventor**. Secured two **"Zhejiang New Talent Program"** projects and participated in the **SRIP** research project. Currently preparing an English paper.

Efficient Path Planning Algorithm Based on Bidirectional A* and Dynamic Threshold Detection — Sep. 2024 – Dec. 2024

- Traditional path search algorithms often face challenges such as low efficiency and inaccuracies when operating in large-scale, complex environments. To overcome these limitations, we proposed an enhanced **bidirectional A*** search algorithm, which utilizes the **Manhattan distance** as a heuristic. This approach resulted in a **14-fold** speedup (from 0.71s to 0.05s) while maintaining a path error of less than **2 pixels**. Furthermore, we integrated an OpenCV-based dynamic color threshold detection method to significantly improve obstacle recognition accuracy and robustness.
- Served as **team leader**; led a 3-member team to complete algorithm design and implementation, earning a **94/100** score and **first place** among undergraduate and graduate teams.

Decay Pruning Method: Pruning Smoothly with a Self-Rectifying Procedure — Jun. 2023 – Jun. 2024

- One-step pruning methods struggle with dynamic network evolution, leading to accuracy loss. We proposed the **Decay Pruning Method (DPM)**, which gradually decays weights during training and uses gradient feedback for smooth, self-rectifying pruning.
- Achieved significant improvements in post-pruning performance and stability. Paper accepted by **ICCV 2025 Workshop** [paper].

Multi-Center MRI Image Classification Based on CNNs — Sep. 2023 – May 2024

- Traditional ML methods rely on manual feature engineering, often missing key information. Multi-center MRI data also face inter-center heterogeneity due to device and protocol variations, limiting generalization. To address this, a **CNN**-based model with automatic feature learning and distribution-alignment strategies was proposed, reducing domain shift and improving cross-center robustness.
- The method enhanced generalization across multiple multi-center datasets, resulting in **two National Invention Patents**.

Internships

Loctek Ergonomic Technology Corp (Algorithm Intern) — Jul. 2025 – Sep. 2025

- Refactored **C++** modules for calibration visualization, extrinsic parameter computation, and point-cloud coordinate transformation into **Python**, improving maintainability and scalability; resolved post-transformation point-cloud misalignment, achieving precise **stereo 3D point cloud** alignment and significantly enhancing spatial consistency and calibration accuracy; stabilized sequential images using **optical flow** and **frame-difference** methods, selecting the frame with minimal camera shake to keep the target centered.

Competitions & Awards

COMAP Mathematical Contest in Modeling | Finalist Winner — Feb. 2024

- Cleaned Wimbledon match data with **SPSS**; built a player evaluation model via **Logistic Regression** and **Entropy-Weighted TOPSIS**; modeled momentum effects with an **LSTM**-based trend predictor to support real-time coaching strategies.
- Led core data analysis and model implementation (**SPSS/Python**); designed the visualization scheme.

COMAP Mathematical Contest in Modeling | Meritorious Winner — Jan. 2025

- Selected eight key indicators via **Pearson correlation**; developed a **Transformer**-based model to forecast medal counts for 200+ countries; quantified the “Star Coach Effect” using **LOESS** trend decomposition and proposed a 3D evaluation framework (focus, breakthroughs, talent import).
- Served as **team leader**; oversaw preprocessing, feature engineering, modeling, visualization, and parts of the paper writing.

Technical Skills

- **Languages:** Python, Java, C/C++, HTML, JavaScript
- **Frameworks/Tools:** PyTorch, OpenCV, Vue.js, MySQL, LaTeX
- **Certifications:** TOPIK-6, TOEIC-LR (740), CET-6 (477), CET-4 (554)